

Day 2: Introduction to Dynamic Routing Theory: What is dynamic routing?
Comparison between static and dynamic routing. Practical -Observe RIP routing updates in a pre-configured network.

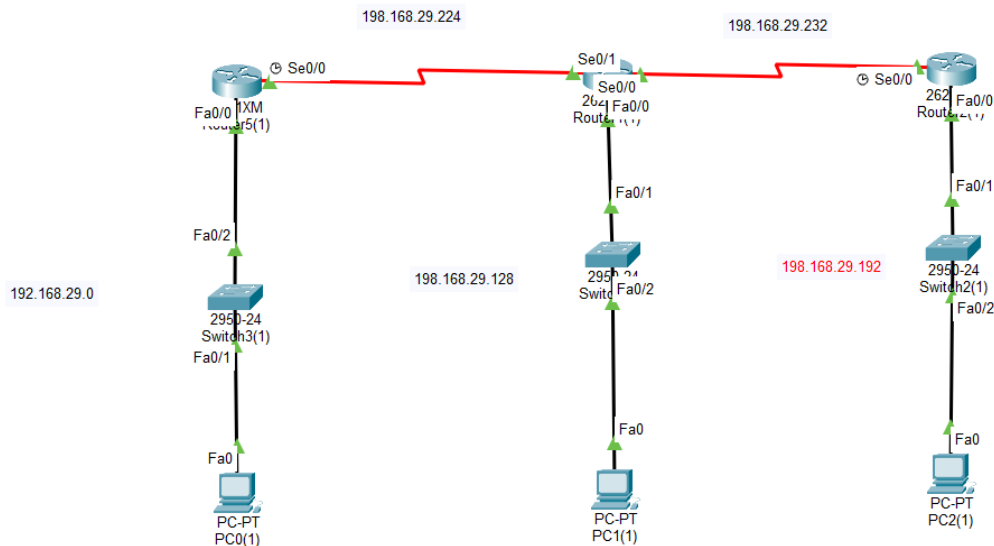
Task 1.

List three differences between static routing and dynamic routing in a table format, using real-world analogies from apps like Google Maps or Swiggy to make each point relatable.

Point	Static Routing	Dynamic Routing
Route selection	Administrator manually configures the route. Analogy: Like manually choosing a fixed road in Google Maps and always following it.	Routers automatically learn and select routes. Analogy: Google Maps automatically changes your route when traffic or a road closure occurs.
Network changes	Does not automatically react to link failures. Analogy: If your fixed Swiggy delivery route is blocked, someone must manually provide another route.	Automatically exchanges routing information and can select another available path. Analogy: Swiggy can dynamically update a delivery path when the original route becomes unavailable.
Scalability	Suitable for small networks because many routes become difficult to maintain manually.	Better for large networks because routers exchange routing information automatically. Analogy: Uber managing thousands of drivers and routes would be impractical if every route had to be manually assigned.

Task 2.

Open Cisco Packet Tracer and load a pre-configured network file with three routers using RIP. Use the 'show ip route' and 'show ip protocols' commands on each router to observe and write down the RIP-learned routes and their metrics.



Router 1: Show IP Protocols & Show IP Route

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.29.0/24 is variably subnetted, 5 subnets, 4 masks
C       192.168.29.0/25 is directly connected, FastEthernet0/0
R       192.168.29.128/26 [120/1] via 192.168.29.230, 00:00:00, Serial0/0
R       192.168.29.192/27 [120/2] via 192.168.29.230, 00:00:00, Serial0/0
C       192.168.29.224/29 is directly connected, Serial0/0
R       192.168.29.232/29 [120/1] via 192.168.29.230, 00:00:00, Serial0/0

Router#show ip protocols
Routing Protocol is "rip"
  Sending updates every 30 seconds, next due in 3 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Redistributing: rip
  Default version control: send version 2, receive 2
    Interface          Send Recv Triggered RIP Key-chain
  FastEthernet0/0      22
  Serial0/0            22
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
  192.168.29.0
Passive Interface(s):
Routing Information Sources:
  Gateway         Distance      Last Update
  192.168.29.230   120          00:00:11
Distance: (default is 120)
Router#
```

Router 2: Show IP Protocols & Show IP Route

```
Router>
Router>
Router>en
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    192.168.29.0/24 is variably subnetted, 5 subnets, 4 masks
R       192.168.29.0/25 [120/1] via 192.168.29.225, 00:00:23, Serial0/1
C       192.168.29.128/26 is directly connected, FastEthernet0/0
R       192.168.29.192/27 [120/1] via 192.168.29.234, 00:00:14, Serial0/0
C       192.168.29.224/29 is directly connected, Serial0/1
C       192.168.29.232/29 is directly connected, Serial0/0

Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 17 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv Triggered RIP Key-chain
  FastEthernet0/0      22
  Serial0/0            22
  Serial0/1            22
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
  192.168.29.0
Passive Interface(s):
Routing Information Sources:
  Gateway         Distance      Last Update
  192.168.29.225   120           00:00:05
  192.168.29.234   120           00:00:22
Distance: (default is 120)
Router#
```

Router 3: Show IP Protocols & Show IP Route

```
Router>
Router>en
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    192.168.29.0/24 is variably subnetted, 5 subnets, 5 masks
R       192.168.29.0/25 [120/2] via 192.168.29.233, 00:00:05, Serial0/0
R       192.168.29.128/26 [120/1] via 192.168.29.233, 00:00:05, Serial0/0
C       192.168.29.192/27 is directly connected, FastEthernet0/0
R       192.168.29.224/29 [120/1] via 192.168.29.233, 00:00:05, Serial0/0
C       192.168.29.232/30 is directly connected, Serial0/0

Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 2 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv Triggered RIP Key-chain
  FastEthernet0/0      22
  Serial0/0            22
Automatic network summarization is in effect
Maximum path: 4
Routing for Networks:
  192.168.29.0
Passive Interface(s):
Routing Information Sources:
  Gateway         Distance      Last Update
  192.168.29.233   120           00:00:11
Distance: (default is 120)
Router#
```

Task 3.

Simulate a network change by shutting down one interface on a router in the RIP-enabled topology. Observe and record how long it takes for the routing tables on the other routers to update and remove the unreachable route.
Hint: Use the 'shutdown' command on an interface and monitor with 'show ip route' repeatedly.

```
Router(config)#int serial 0/1
%Invalid interface type and number
Router(config)#int serial 0/0
Router(config-if)#shutdown
```

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.168.29.1    YES manual up          up
FastEthernet0/1          unassigned      YES unset  down        down
Serial0/0                 192.168.29.225  YES manual administratively down down
Router#
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.29.0/25 is subnetted, 1 subnets
C       192.168.29.0 is directly connected, FastEthernet0/0
Router#
```

RIP Timer	Default
Update	30 seconds
Invalid	180 seconds
Hold-down	180 seconds
Flush	240 seconds

Task 4.

Explain in 3-4 sentences how dynamic routing in apps like Uber or Zomato delivery tracking is similar to RIP dynamic routing in networks. Focus on how routes or paths are updated automatically in both cases.

- Dynamic routing in Uber or Zomato delivery tracking is similar to RIP because both continuously update information about available paths.
- In RIP, routers exchange routing information and update their routing tables when network paths change.
- Similarly, Uber or Zomato can update a driver's or delivery partner's route when traffic, road closures, or other conditions change.
- In both cases, the goal is to automatically adapt to changing paths rather than requiring every route to be manually configured.